

E026

AI for IT Automation and Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A university security team wants to spot unusual data flows from lab servers. It has no labeled examples of past incidents on these systems. The servers were bought with a federal research grant and run a popular Linux distribution. Which approach fits?

- A. Reinforcement learning, because it is the most advanced.
- B. A rule for each Linux version.
- C. Unsupervised anomaly detection.
- D. Supervised classification.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A team has 50,000 emails already labeled as phishing or legitimate and wants to sort new emails into those two groups. Which approach fits?

- A. Regression.
- B. No model; blocklists catch all phishing.
- C. Supervised classification.
- D. Unsupervised clustering.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A SOC that works three shifts lowers its anomaly model's alert threshold after missing an intrusion last quarter. The model was bought from a vendor based in Austin. What should the SOC expect?

- A. Fewer missed threats and fewer false positives.
- B. No change, since the vendor tuned it.
- C. More missed threats and fewer false positives.
- D. Fewer missed threats but more false positives.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Analysts are overwhelmed by false alarms, so the team raises the alert threshold. Which new risk must it watch?

- A.** The model stops producing any alerts.
- B.** More real threats go undetected.
- C.** Even more false alarms.
- D.** Slower network speed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A bank sees about 100,010 logins a day; about 10 are from compromised accounts. Its detector catches 90% of compromised logins and wrongly flags 1% of normal ones. The detector runs on a new cloud cluster. About how many alerts fire per day, and what share are real?

- A.** About 1,000 alerts; 99% are real.
- B.** About 9 alerts; all real.
- C.** About 1,009 alerts; under 1% are real.
- D.** About 1,009 alerts; 90% are real.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A SOC can review at most 200 alerts a day. It sees 50,000 normal events a day and very few real threats. What is the highest false-positive rate the detector can have? (Ignore true alerts.)

- A.** 4%
- B.** 0%, since any false positive is unacceptable.
- C.** 250%
- D.** 0.4%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A security team at a 24-hour hospital is building an automated playbook. Proposed steps: enrich alerts with threat-intelligence data, open a ticket, and automatically shut down any server with a suspicious process—including the patient-monitoring system. The team uses a well-reviewed SOAR platform. What should change?

- A. Switch to a different SOAR platform.
- B. Keep enrichment and ticketing automatic; require human approval before shutting down critical systems.
- C. Automate nothing; do all steps by hand.
- D. Automate everything, since speed always wins.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which step is the best candidate for full automation in a security playbook?

- A. Looking up an IP address's reputation and attaching it to the ticket.
- B. Deciding whether to notify regulators.
- C. Wiping a production database server.
- D. Disabling the CEO's account.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

When a core network switch fails at a retailer's data center, 1,400 alerts fire from servers, apps, and monitors within ten minutes. The on-call engineer is on her first week. What does an AIOps platform add here?

- A. Replaces the failed switch automatically.
- B. Correlates the alerts into one incident and points to the switch as the likely root cause.
- C. Assigns a more senior engineer.
- D. Generates more alerts for better coverage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A traffic-anomaly model worked well for a year. After the company moved most apps to a new cloud platform, false positives tripled. The model is written in Python and runs every five minutes. What is the most likely cause and fix?

- A.** Nothing; models never degrade once validated.
- B.** Run the model less often.
- C.** The code language; rewrite it.
- D.** Model drift; retrain on current traffic and keep monitoring performance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A company trains its malware classifier on files collected from a public sharing site that anyone can upload to. The data science team has six members and a large GPU budget. What is the main risk to the model?

- A.** Poisoned training data; control data sources, verify provenance, and validate before training.
- B.** None; security AI is secure by design.
- C.** The team is too small.
- D.** Not enough GPUs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

An accountant who always logs in from one city between 8 a.m. and 6 p.m. logs in with valid credentials at 3 a.m. and downloads 40 times her normal data volume. She has worked at the firm for 11 years. Which capability flags this best?

- A.** User and entity behavior analytics comparing activity to her baseline.
- B.** Password complexity rules.
- C.** A single company-wide download limit.
- D.** Nothing; long-tenured staff are trusted.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A university security team wants to spot unusual data flows from lab servers. It has no labeled examples of past incidents on these systems. ~~The servers were bought with a federal research grant and run a popular Linux distribution.~~ Which approach fits?

Target word: learning approach **Decides:** No labels exist; the goal is unusual behavior.

A	Adjacent category	Learns by reward in an environment; not suited to this detection task.
B	Camouflage	The OS does not decide the approach.
C	Correct	Finds deviations without labeled examples.
D	Inverted operation	Needs labeled incidents, which do not exist.

2. Answer C Concept 1 • transfer

A team has 50,000 emails already labeled as phishing or legitimate and wants to sort new emails into those two groups. Which approach fits?

Target word: reverse: labeled data exists **Decides:** Labeled classes → supervised.

A	Confusion set	Predicts a number, not a category.
B	Absolute language	Blocklists miss new senders.
C	Correct	Labeled examples train a classifier.
D	Inverted operation	Ignores the labels the team already has.

3. Answer D Concept 2

A SOC ~~that works three shifts~~ lowers its anomaly model's alert threshold after missing an intrusion last quarter. ~~The model was bought from a vendor based in Austin.~~ What should the SOC expect?

Target word: effect of lowering threshold **Decides:** Lower threshold → more alerts.

A	Absolute language	One threshold cannot cut both errors.
B	Camouflage	The vendor does not decide the trade-off.
C	Inverted operation	That is what raising the threshold does.
D	Correct	Sensitivity rises; so do false alarms.

4. Answer B Concept 2 • transfer

Analysts are overwhelmed by false alarms, so the team raises the alert threshold. Which new risk must it watch?

Target word: reverse: raise threshold **Decides:** Raising it trades false positives for false negatives.

A	Absolute language	Alerts fall, not to zero.
B	Correct	False negatives rise.
C	Inverted operation	Raising the threshold reduces them.
D	Adjacent category	Threshold changes do not affect throughput.

5. Answer C Concept 3

A bank sees about 100,010 logins a day; about 10 are from compromised accounts. Its detector catches 90% of compromised logins and wrongly flags 1% of normal ones. ~~The detector runs on a new cloud cluster.~~ About how many alerts fire per day, and what share are real?

Target word: alert volume and precision **Decides:** 9 true alerts vs. 1% of 100,000 normal logins.

A	Inverted operation	Reads a 1% false-positive rate as 99% precision.
B	Dropped element	Leaves out the false positives.
C	Correct	9 true + 1,000 false; $9 \div 1,009 \approx 0.9\%$.
D	Confusion set	Confuses detection rate with the share of alerts that are real.

6. Answer D Concept 3 · transfer

A SOC can review at most 200 alerts a day. It sees 50,000 normal events a day and very few real threats. What is the highest false-positive rate the detector can have? (Ignore true alerts.)

Target word: solve for maximum false-positive rate **Decides:** $200 \div 50,000$.

A	Wrong stem number	Off by a factor of ten; would produce 2,000 alerts.
B	Absolute language	Some false positives are expected; capacity sets the limit.
C	Inverted operation	Divides events by capacity.
D	Correct	$200 \div 50,000 = 0.004$.

7. Answer B Concept 4

A security team at a 24-hour hospital is building an automated playbook. Proposed steps: enrich alerts with threat-intelligence data, open a ticket, and automatically shut down any server with a suspicious process—including the patient-monitoring system. ~~The team uses a well-reviewed SOAR platform.~~ What should change?

Target word: where to keep a human **Decides:** Shutdown of critical systems is high-impact.

A	Camouflage	The tool is not the issue.
B	Correct	Automate low-risk steps; gate disruptive ones.
C	Inverted operation	Wastes analyst time on routine steps.
D	Absolute language	Some actions carry more risk than delay.

8. Answer A Concept 4 · transfer

Which step is the best candidate for full automation in a security playbook?

Target word: reverse: pick the safe step **Decides:** Repetitive, low-risk, well-defined.

A	Correct	Repeatable, low-risk enrichment.
B	Confusion set	A legal judgment, not a routine task.
C	Adjacent category	Irreversible and disruptive.
D	Inverted operation	High-impact; needs human judgment.

9. Answer B Concept 5

When a core network switch fails at a retailer's data center, 1,400 alerts fire from servers, apps, and monitors within ten minutes. ~~The on-call engineer is on her first week.~~ What does an AIOps platform add here?

Target word: AIOps value **Decides:** Many symptoms, one cause.

A	Adjacent category	Software cannot replace hardware.
B	Correct	Noise reduction and root-cause hints.
C	Camouflage	Staffing is not what AIOps does.
D	Inverted operation	More noise slows response.

10. Answer D Concept 6

A traffic-anomaly model worked well for a year. After the company moved most apps to a new cloud platform, false positives tripled. ~~The model is written in Python and runs every five minutes.~~ What is the most likely cause and fix?

Target word: cause of degradation **Decides:** The environment changed after training.

A	Absolute language	Performance changes with the environment.
B	Dropped element	Does not address the mismatch.
C	Camouflage	Language does not cause drift.
D	Correct	New patterns no longer match training data.

11. Answer A Concept 7

A company trains its malware classifier on files collected from a public sharing site that anyone can upload to. ~~The data science team has six members and a large GPU budget.~~ What is the main risk to the model?

Target word: AI-specific risk **Decides:** Untrusted training data.

A	Correct	Untrusted inputs can skew the model.
B	Absolute language	AI systems have their own attack surface.
C	Camouflage	Team size is not the risk.
D	Adjacent category	Compute capacity does not address data integrity.

12. Answer A Concept 8

An accountant who always logs in from one city between 8 a.m. and 6 p.m. logs in with valid credentials at 3 a.m. and downloads 40 times her normal data volume. ~~She has worked at the firm for 11 years.~~ Which capability flags this best?

Target word: detection capability **Decides:** Valid login but departure from her own normal.

A	Correct	Flags deviation from individual norms.
B	Adjacent category	Her credentials were valid.
C	Confusion set	Ignores individual baselines.
D	Camouflage	Tenure does not decide risk.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Supervised vs. unsupervised detection	Supervised models learn from labeled examples and classify known patterns; unsupervised models find anomalies without labels and can flag novel behavior.	Expecting a supervised model trained on past labels to recognize a never-seen pattern.	Whether labeled examples exist for what must be detected.
2	Threshold trade-off	Lowering an alert threshold catches more true threats (fewer false negatives) but raises false positives; raising it does the reverse.	Believing a threshold change can cut both error types at once.	A threshold is moved up or down.
3	Base-rate effect	When real threats are rare, even a small false-positive rate produces far more false alarms than true alerts.	Reading a 1% false-positive rate as “99% of alerts are real.”	A rare event count, a detection rate, and a false-positive rate.
4	Automation with human oversight (SOAR)	Automate repetitive, well-defined, low-risk steps (enrichment, ticketing); keep a human approval step for high-impact or irreversible actions.	Fully automating disruptive actions such as shutting down production systems.	An action with major business impact in an automated playbook.
5	AIOps event correlation	AIOps correlates related alerts into incidents and highlights probable root cause, reducing noise so teams act faster.	Treating more alerts as better monitoring.	An alert storm from one underlying issue.
6	Model drift	Model performance degrades as the environment changes; monitor performance and retrain on current data.	Assuming a model that tested well at deployment stays accurate.	Performance falling after a change in systems or behavior.
7	Securing the AI itself	AI systems face threats such as poisoned training data and manipulated inputs; defenses include data provenance, access control on training pipelines, validation, and monitoring.	Assuming AI tools are inherently secure because they are security products.	Questions about where training data came from or who can change it.
8	Behavioral baselines (UEBA)	User and entity behavior analytics compares activity to each entity’s own normal baseline, so unusual behavior stands out even with valid credentials.	Using one global rule for all users regardless of their normal patterns.	Valid login, unusual pattern.

Course scope (public description): AI for IT Automation and Security explores the intersection of AI and network security. It emphasizes how AI techniques can enhance security practices, automate tasks, optimize a network, and improve threat detection and response.